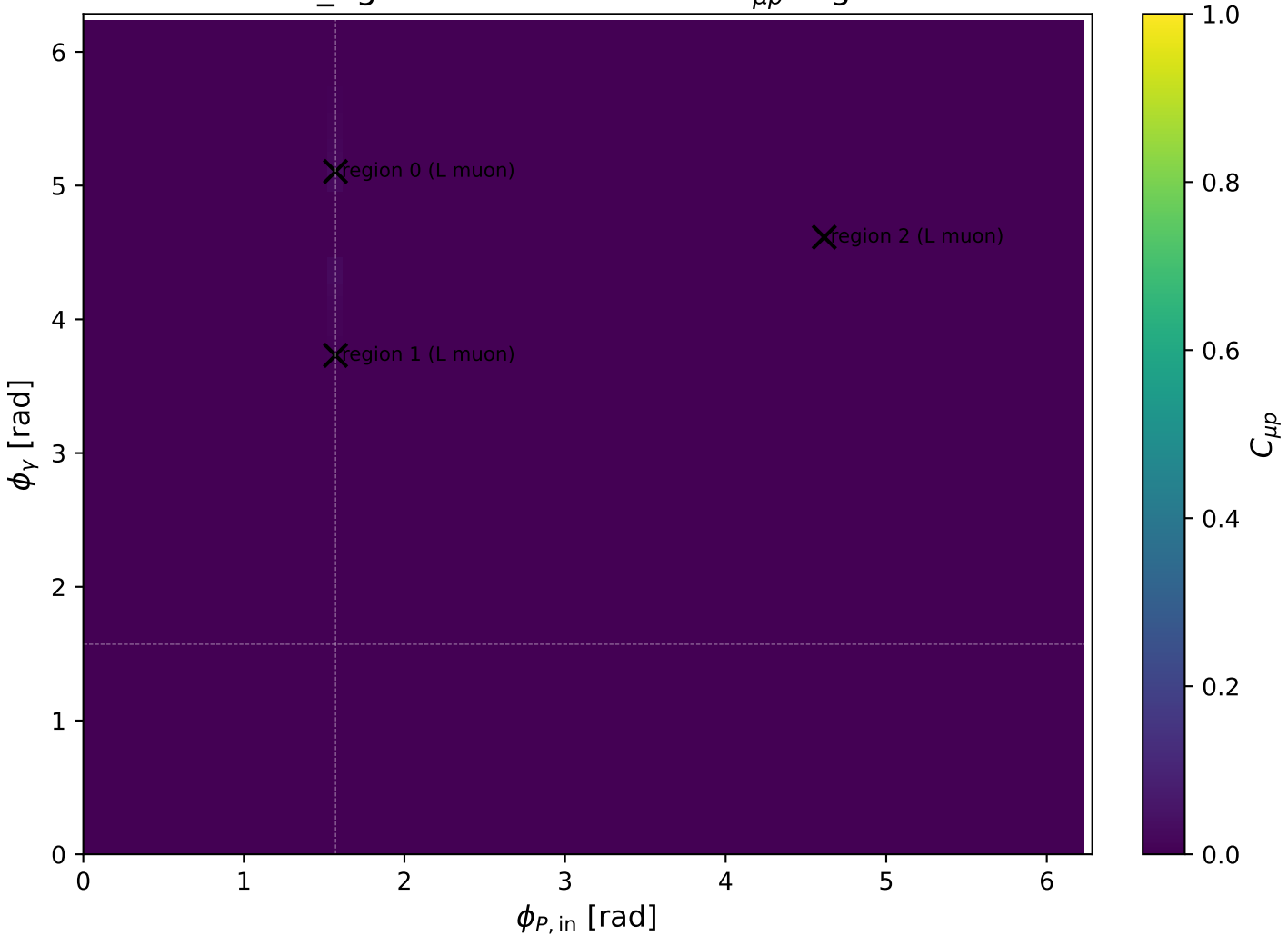
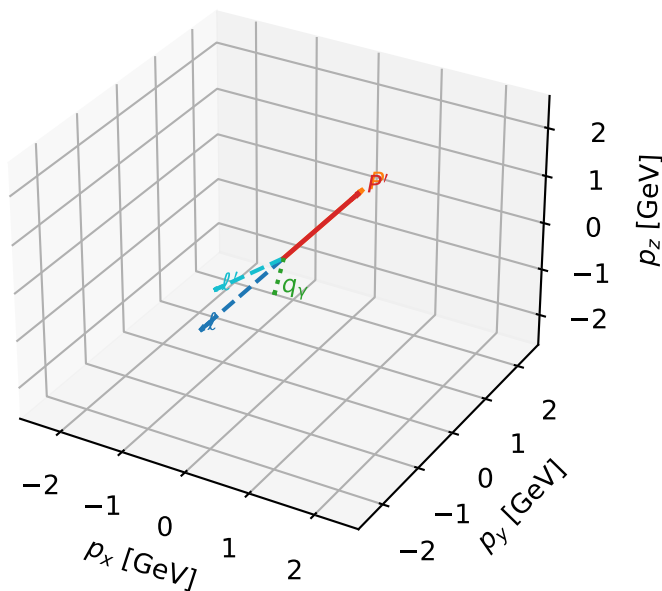


mid\_Egamma L muon: max  $C_{\mu p}$  regions



# L muon characteristic max $C_{\mu p}$ configuration

3D momenta



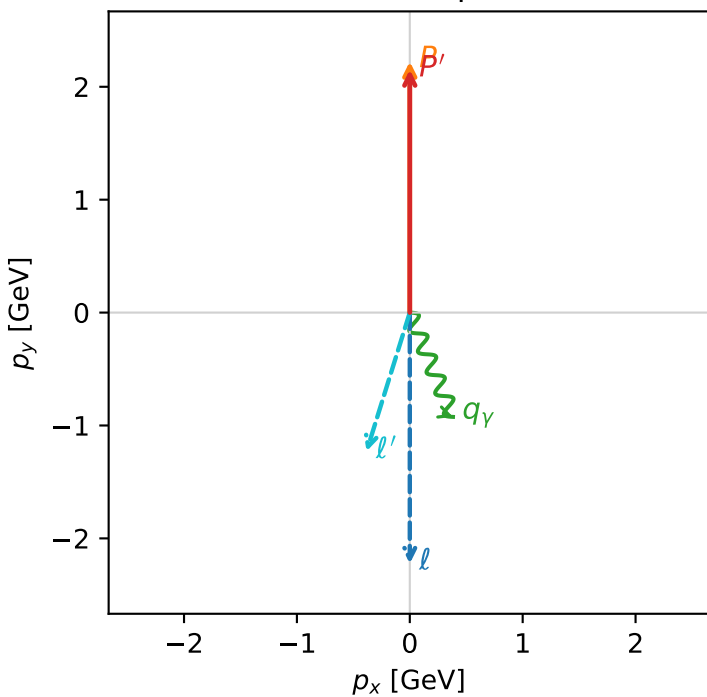
c\_ep\_mid\_egamma\_l\_electron\_region\_0  $C_{\mu p}$ =0.0242163  
 region: mid\_Egamma\_L\_electron\_region\_0 final-pair delta\_xy=2.83956 rad  
 outgoing-state purity: 0.609605  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.15215$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=1$ ,  $\phi_\gamma=5.10509$   
 $Q^2=0.262333$ ,  $x_B=0.0293623$ ,  $t=-0.000782118$ ,  $W^2=9.55186$ ,  $y=0.433184$   
 $\theta(\ell', \gamma)=39.8051$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|            |      |        |        |         |        |         |        |         |
|------------|------|--------|--------|---------|--------|---------|--------|---------|
| $\ell$     | $E=$ | 2.2256 | $p_x=$ | -0.0000 | $p_y=$ | -2.2231 | $p_z=$ | -0.0000 |
| $P$        | $E=$ | 2.4129 | $p_x=$ | 0.0000  | $p_y=$ | 2.2231  | $p_z=$ | 0.0000  |
| $\ell'$    | $E=$ | 1.2908 | $p_x=$ | -0.3827 | $p_y=$ | -1.2283 | $p_z=$ | 0.0000  |
| $P'$       | $E=$ | 2.3477 | $p_x=$ | 0.0000  | $p_y=$ | 2.1522  | $p_z=$ | 0.0000  |
| $q_\gamma$ | $E=$ | 1.0000 | $p_x=$ | 0.3827  | $p_y=$ | -0.9239 | $p_z=$ | 0.0000  |

Transverse plane



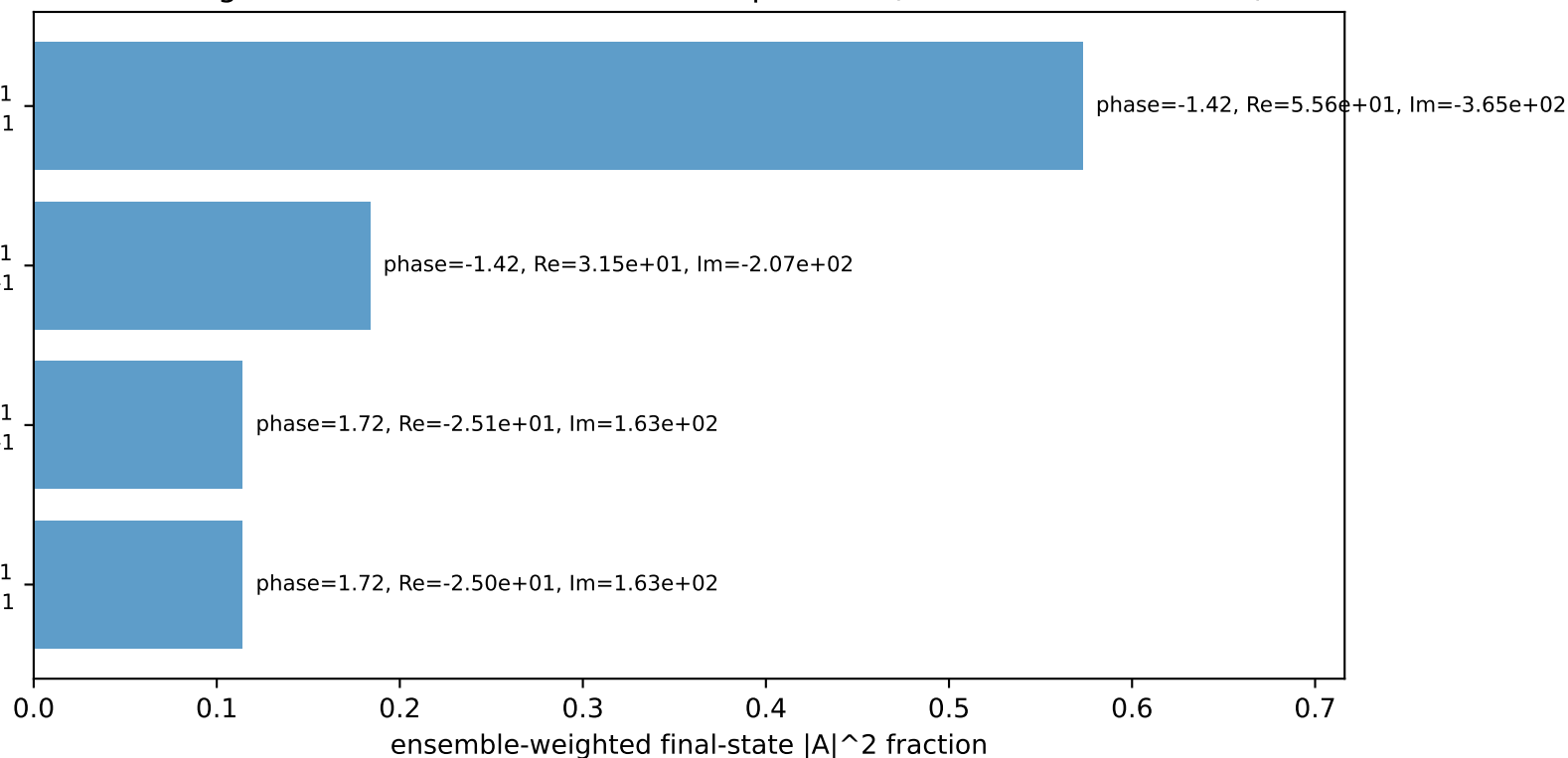
$h_e=+1, h_p=-1, h_\gamma=+1$   
 muon L, proton h=+1

$h_e=+1, h_p=+1, h_\gamma=-1$   
 muon L, proton h=-1

$h_e=+1, h_p=-1, h_\gamma=-1$   
 muon L, proton h=-1

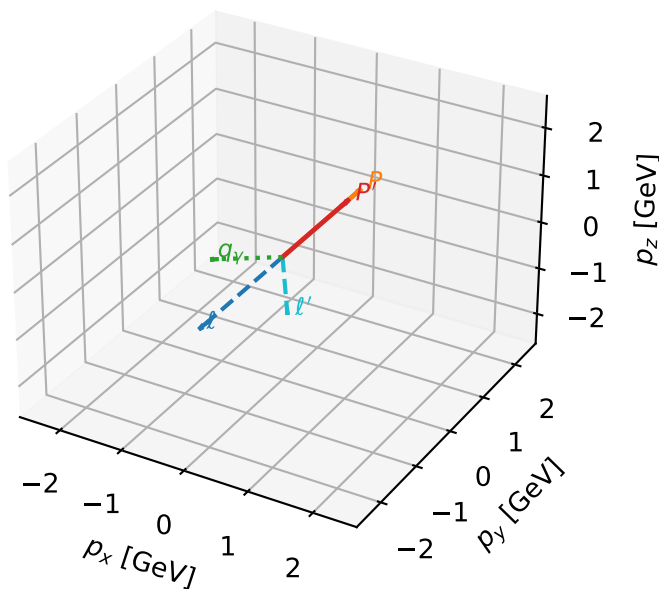
$h_e=+1, h_p=+1, h_\gamma=-1$   
 muon L, proton h=+1

Leading initial-ensemble/final-state components (N=4, retained=98.4%)



# L muon characteristic max $C_{\mu p}$ configuration

3D momenta



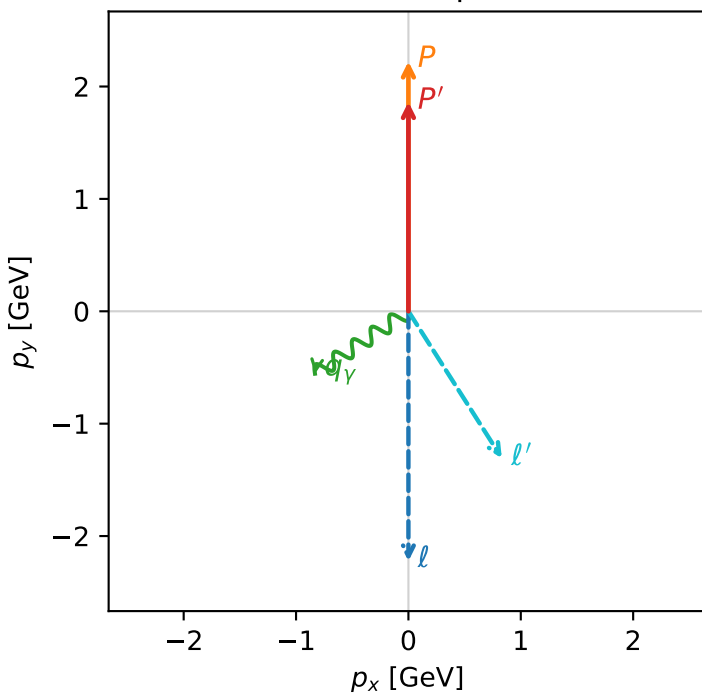
c\_ep\_mid\_egamma\_l\_electron\_region\_1  $C_{\{\mu p\}}=0.0048206$   
region: mid\_Egamma\_L\_electron\_region\_1 final-pair delta\_xy=2.57509 rad  
outgoing-state purity: 0.59059  
kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=1.86282$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_P=1.5708$   
 $E_\gamma=1$ ,  $\phi_\gamma=3.73064$   
 $Q^2=1.07755$ ,  $x_B=0.147233$ ,  $t=-0.0227212$ ,  $W^2=7.12099$ ,  $y=0.354849$   
 $\theta(\ell', \gamma)=88.7082$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2256$   $p_x= -0.0000$   $p_y= -2.2231$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 0.0000$   $p_y= 2.2231$   $p_z= 0.0000$   
 $\ell'$   $E= 1.5529$   $p_x= 0.8315$   $p_y= -1.3072$   $p_z= 0.0000$   
 $P'$   $E= 2.0856$   $p_x= 0.0000$   $p_y= 1.8628$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= -0.8315$   $p_y= -0.5556$   $p_z= 0.0000$

Transverse plane



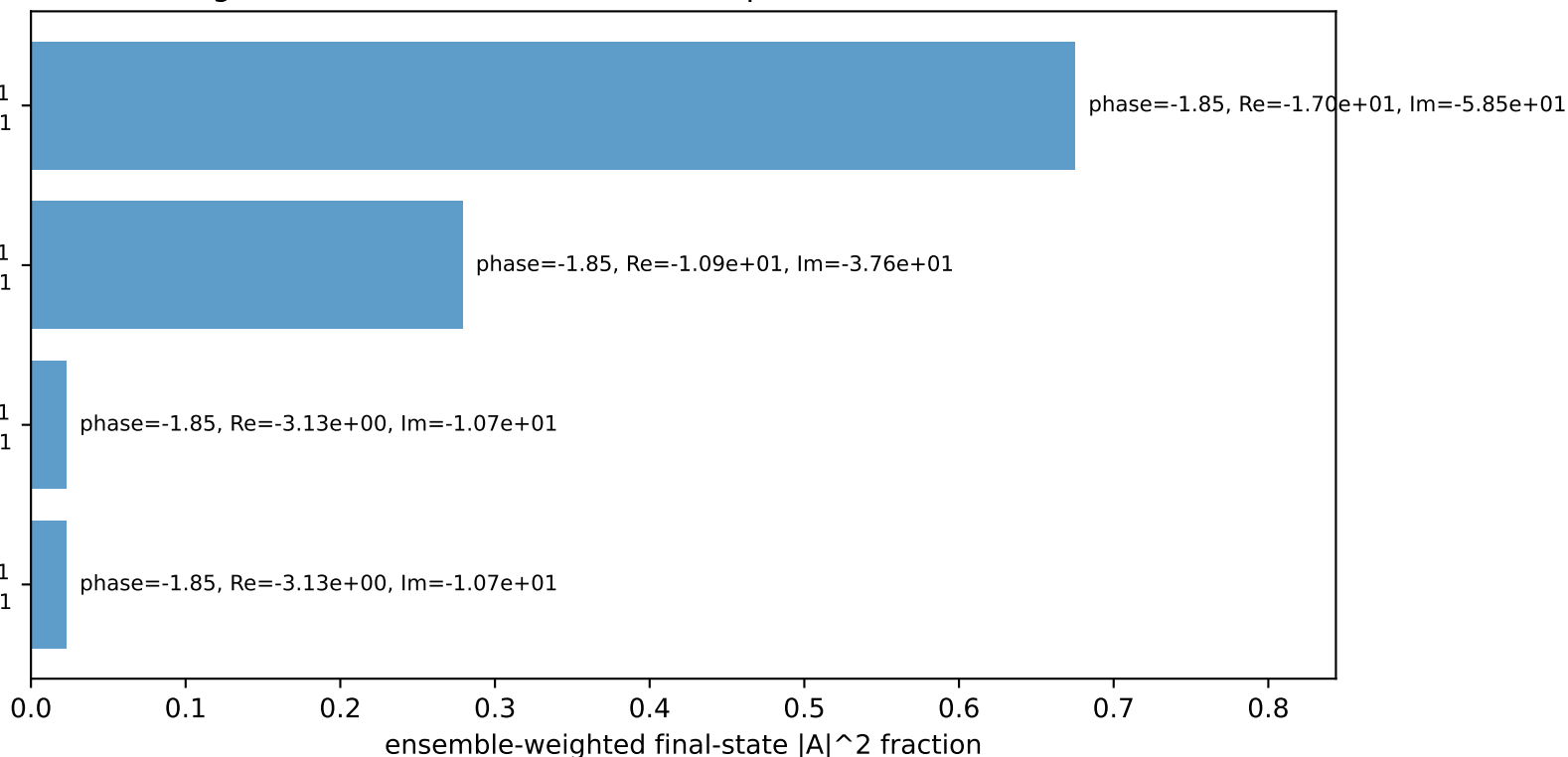
$h_e=+1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
muon L, proton h=+1

$h_e=+1$ ,  $h_p=+1$ ,  $h_\gamma=-1$   
muon L, proton h=-1

$h_e=+1$ ,  $h_p=-1$ ,  $h_\gamma=-1$   
muon L, proton h=-1

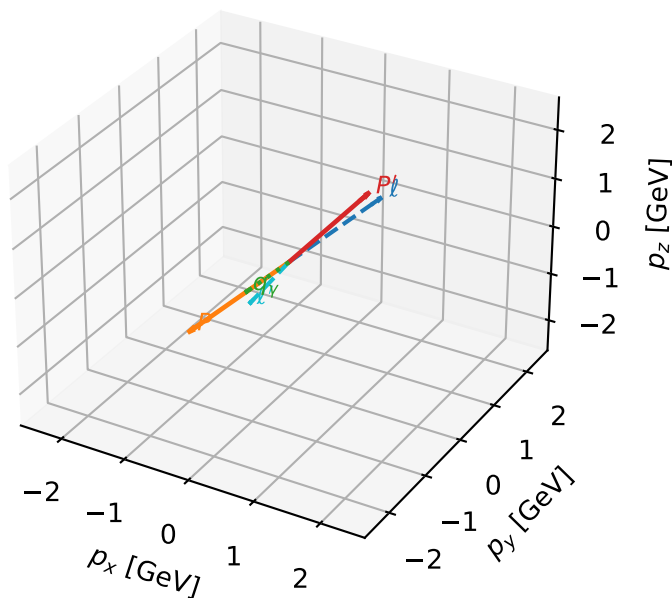
$h_e=+1$ ,  $h_p=+1$ ,  $h_\gamma=-1$   
muon L, proton h=+1

Leading initial-ensemble/final-state components (N=4, retained=99.9%)



# L muon characteristic max $C_{\mu p}$ configuration

3D momenta

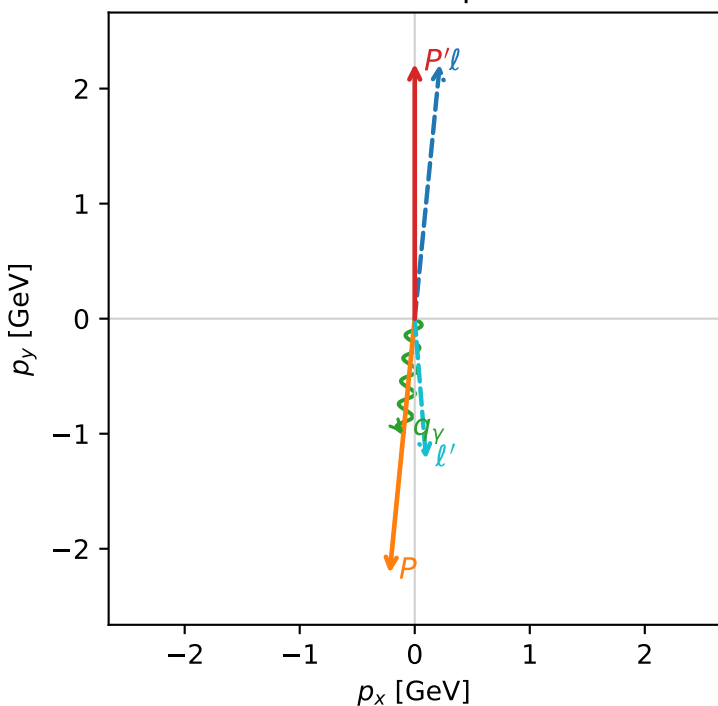


c\_ep\_mid\_egamma\_l\_electron\_region\_2  $C_{\mu p}=0.00169423$   
region: mid\_Egamma\_L\_electron\_region\_2 final-pair delta\_xy=3.06157 rad  
outgoing-state purity: 0.999949  
kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.2175$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=1.47262$ ,  $\phi_p=4.61421$   
 $E_Y=1$ ,  $\phi_Y=4.61421$   
 $Q^2=10.822$ ,  $x_B=0.539721$ ,  $t=-19.6716$ ,  $W^2=10.109$ ,  $y=0.972184$   
 $\theta(\ell', \gamma)=10.2097$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2256$   $p_x= 0.2179$   $p_y= 2.2124$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -0.2179$   $p_y= -2.2124$   $p_z= 0.0000$   
 $\ell'$   $E= 1.2308$   $p_x= 0.0980$   $p_y= -1.2223$   $p_z= 0.0000$   
 $P'$   $E= 2.4077$   $p_x= 0.0000$   $p_y= 2.2175$   $p_z= 0.0000$   
 $q_Y$   $E= 1.0000$   $p_x= -0.0980$   $p_y= -0.9952$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=100.0%)

